

DATABASE MANAGEMENT SYSTEMS (CS301)

5 Marks Question Bank

UNIT 1 – Introduction to Databases

1. Define DBMS and explain its advantages.
2. Differentiate File System and DBMS.
3. Explain the three-level architecture of DBMS.
4. What is Data Independence?
5. Explain Logical and Physical Data Independence.
6. Discuss applications of DBMS.

UNIT 2 – ER Model & Relational Model

1. Define Entity and Attribute.
2. Differentiate Strong Entity and Weak Entity.
3. Explain Cardinality Constraints.
4. What is a Relational Model?
5. Define Primary Key and Foreign Key.
6. Explain Referential Integrity.

UNIT 3 – SQL Queries

1. Explain DDL commands with examples.
2. Explain DML commands with examples.
3. What is a SELECT query?
4. Differentiate INNER JOIN and OUTER JOIN.
5. What are Nested Queries?
6. Explain Stored Procedures.

UNIT 4 – Functional Dependencies & Normalization

1. Define Functional Dependency.
2. Explain Full Functional Dependency.
3. What is Partial Dependency?
4. Explain Transitive Dependency.
5. Define 1NF, 2NF and 3NF.
6. What is BCNF?

UNIT 5 – Transactions

1. Define Transaction in DBMS.
2. Explain ACID Properties.
3. What is Concurrency Control?
4. Differentiate Shared Lock and Exclusive Lock.
5. What is Deadlock?
6. Explain Recovery Concepts.

Exam Preparation Tips

Focus on DBMS architecture, ER diagrams, keys, SQL commands, joins, normalization, functional dependencies, ACID properties, and concurrency control as these are frequently asked 5-mark questions.